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Inventor(s)

NISHIKOORI; Nobuharu et al.

INDUCTION HEATING COOKER

Abstract

An induction heating cooker including a top plate; a substrate positioned on a bottom surface of the top plate; at least one heating coil on the substrate and configured to inductively heat a heating object, the at least one heating coil being sheet-shaped; and at least one sensor coil configured to detect a position of the heating object, wherein the at least one sensor coil is on the substrate so that the at least one heating coil and the at least one sensor coil are integrated into a single body.

Inventors: NISHIKOORI; Nobuharu (Yokohama-shi, JP), SODA; Yasushi (Yokohama-shi, JP), OOSAKO; Seisaku (Yokohama-shi, JP), KANAGAWA; Tomoyuki (Yokohama-shi, JP)

Applicant: Samsung Electronics Co., Ltd. (Suwon-si, KR)

Family ID: 91485141

Assignee: Samsung Electronics Co., Ltd. (Suwon-si, KR)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation application of International Application No. PCT/KR2023/018756, filed on Nov. 21, 2023, which is based on and claims the benefit of Japanese Patent Application Number 2022-201363, filed on Dec. 16, 2022, and Japanese Patent Application Number 2023-183791, filed on Oct. 26, 2023, the disclosures of which are incorporated by reference herein in their entireties.

TECHNICAL FIELD

[0002] The present disclosure relates to an induction heating cooker for inductively heating a heating object such as a pot.

BACKGROUND ART

[0003] Conventional induction heating cookers use litz wires in the shape of a coil as a heating coil for heating a heating object placed on a top plate. In order to heat a heating object placed at an arbitrary position on the top plate, the induction heating cookers include a sensor coil to detect a position of the heating object. The heating coil and the sensor coil are arranged as different structures. This type of induction heating cooker is disclosed in Japanese Patent Application No. 2016-77445.

DISCLOSURE OF INVENTION

Solution to Problem

[0004] Aspects of embodiments of the disclosure will be set forth in part in the description which follows and, in part, will be apparent from the description, or may be learned by practice of the presented embodiments.

[0005] According to an embodiment of the disclosure, an induction heating cooker includes a top plate; a substrate positioned on a bottom surface of the top plate; at least one heating coil on the substrate and configured to inductively heat a heating object, the at least one heating coil being sheet-shaped; and at least one sensor coil configured to detect a position of the heating object, wherein the at least one sensor coil is on the substrate so that the at least one heating coil and the at least one sensor coil are integrated into a single body.

[0006] According to an embodiment of the disclosure, the at least one sensor coil may include at least two sensor coils arranged for the at least one heating coil. The at least two sensor coils may be arranged diagonally about a center of the at least one heating coil.

[0007] According to an embodiment of the disclosure, each sensor coil of the at least two sensor coils may be spaced apart from a corner portion of the at least one heating coil toward the center of the at least one heating coil.

[0008] According to an embodiment of the disclosure, each sensor coil of the at least two sensor coils may be positioned between a corner portion of the at least one heating coil and two orthogonal lines that pass through the center of the at least one heating coil.

[0009] According to an embodiment of the disclosure, the at least one heating coil may include at least two heating coils arranged on the substrate. A distance between the at least two heating coils may be smaller than half an outer diameter of each heating coil of the at least two heating coils.

[0010] According to an embodiment of the disclosure, the at least one heating coil may include at least two heating coils arranged on the substrate. The substrate may include two or more heating coil layers each including at least one heating coil of the at least two heating coils, the two or more

heating coil layers being insulated from each other, a sensor layer including the at least one sensor coil, and an insulation layer positioned between the two or more heating coil layers and the sensor layer. The sensor layer may be positioned closer to the top plate than are the two or more heating coil layers.

[0011] According to an embodiment of the disclosure, one heating coil layer of the two or more heating coil layers may be connected to a resonance capacitor, and may be positioned farther away from the sensor layer than is another heating coil layer of the two or more heating coil layers.

[0012] According to an embodiment of the disclosure, the induction heating cooker may further include a temperature sensor configured to detect a temperature of at least one of the at least one heating coil or the heating object, and integrated into the single body.

[0013] According to an embodiment of the disclosure, the temperature sensor may be positioned at a center of the at least one heating coil, near the center of the at least one heating coil, or near a connection connector to which the at least one heating coil is connected.

[0014] According to an embodiment of the disclosure, the induction heating cooker may further include a control board to control the at least one heating coil; and a connection connector connecting the at least one heating coil to the control board, and provided on the substrate. The connection connector may be mounted on the substrate in a direction opposite to the top plate.

[0015] According to an embodiment of the disclosure, the connection connector and the at least one heating coil may be on a heating coil layer of the substrate.

[0016] According to an embodiment of the disclosure, the induction heating cooker may further include a driving circuit portion connected to the at least one sensor coil; and at least one protection circuit portion positioned respectively between the at least one sensor coil and the driving circuit portion, and configured to protect the driving circuit portion from an induced voltage generated in the at least one sensor coil.

[0017] According to an embodiment of the disclosure, the at least one sensor coil may include a plurality of sensor coils. The at least one protection circuit portion may include a plurality of protection circuit portions. The plurality of sensor coils may be connected to the driving circuit portion. One protection circuit portion of the plurality of protection circuit portions may be provided for each sensor coil of the plurality of sensor coils.

[0018] According to an embodiment of the disclosure, the induction heating cooker may further include a temperature sensor configured to detect a temperature of at least one of the at least one heating coil or the heating object. The induction heating cooker may be configured to correct an output value of the at least one sensor coil by using the temperature detected by the temperature sensor.

[0019] According to an embodiment of the disclosure, the induction heating cooker may be configured to compensate for a change of an output value of the at least one sensor coil caused due to magnetic flux absorption by the at least one heating coil when the at least one heating coil starts to operate.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0020] The above and other aspects, features, and advantages of certain embodiments of the disclosure will be more apparent from the following description, taken in conjunction with the accompanying drawings listed below.

[0021] FIG. 1 is a schematic diagram showing an entire configuration of an induction heating cooker according to an embodiment of the present disclosure.

[0022] FIG. 2 is a schematic diagram of an embodiment of a position detection sensor.

[0023] FIG. 3 is a schematic diagram of an embodiment of a substrate.

[0024] FIG. 4 is a schematic diagram of an embodiment of a substrate.

[0025] FIG. 5 is a functional block diagram of an embodiment of a control device.

[0026] FIG. 6 is a view for describing arrangements of sensor coils with respect to heating coils.

[0027] FIG. 7 is a schematic diagram showing an embodiment of an arrangement of sensor coils with respect to heating coils.

[0028] FIG. 8 shows calculation results of eddy current loss according to arrangements of sensor coils.

[0029] FIG. 9 is a schematic diagram showing an embodiment of an arrangement of sensor coils.

[0030] FIG. 10 is a schematic diagram for describing a configuration of a substrate in the case of an existing heating coil using litz wires.

[0031] FIG. 11 is a schematic diagram showing a configuration of a substrate in the case of using a heating coil according to an embodiment of the present disclosure.

[0032] FIG. 12 is a circuit diagram showing a configuration of a sensor coil, a driving circuit portion, a MUX portion, and a protection circuit portion according to an embodiment of the present disclosure.

[0033] FIG. 13 schematically shows operation states of an embodiment of a protection circuit portion.

[0034] FIG. 14 shows a count value when no temperature correction is performed and a count value when temperature correction is performed.

[0035] FIG. 15 shows a count value when magnetic flux absorption is not corrected and a count value when magnetic flux absorption is corrected.

MODE FOR THE INVENTION

[0036] Various embodiments of the present disclosure and terms used therein are not intended to limit the technical features described in the present disclosure to particular embodiments, and it should be construed as including various modifications, equivalents, or alternatives of a corresponding embodiment.

[0037] With regard to description of drawings, similar reference numerals may be used for similar or related components.

[0038] A singular form of a noun corresponding to an item may include one item or a plurality of the items unless context clearly indicates otherwise.

[0039] As used herein, each of the expressions “A or B,” “at least one of A and B,” “at least one of A or B,” “A, B, or C,” “at least one of A, B, and C,” and “at least one of A, B, or C,” may include one or all possible combinations of the items listed together with a corresponding expression among the expressions.

[0040] The term “and/or” includes any and all combinations of one or more of a plurality of associated listed items.

[0041] The terms “1.sup.st”, “2.sup.nd”, “first”, “second”, etc., may be used only to distinguish one component from another, not intended to limit the corresponding component in other aspects (e.g., importance or order).

[0042] It is said that one (e.g., first) component is “coupled” or “connected” to another (e.g., second) component, with or without the terms “functionally” or “communicatively”. When referenced, it means that the one component can be connected to the other component directly (e.g., by wire), wirelessly, or through a third component.

[0043] It will be understood that when the terms “includes,” “comprises,” “including,” and/or “comprising,” when used in this specification, specify the presence of stated features, figures, steps, operations, components, members, or combinations thereof, but do not preclude the presence or addition of one or more other features, figures, steps, operations, components, members, or combinations thereof.

[0044] An expression that one component is “connected”, “coupled”, “supported”, or “in contact” with another component includes a case in which the components are directly “connected”,

“coupled”, “supported”, or “in contact” with each other and a case in which the components are indirectly “connected”, “coupled”, “supported”, or “in contact” with each other through a third component.

[0045] It will also be understood that when one component is referred to as being “on” or “over” another component, it can be directly on the other component or intervening components may also be present.

[0046] In an induction heating cooker, in the case in which a heating coil and a sensor coil are different structures, a misalignment in position of the sensor coil with respect to the heating coil may occur, which may make it difficult to detect a slight difference in eddy current flowing through a heating object or an exact position of the heating object. Therefore, accuracy in detecting the position of the heating object may deteriorate. In order to ensure the accuracy of position detection, the number of sensor coils needs to increase. In the case of using litz wires as a heating coil, a member for maintaining the litz wires in the shape of a coil is needed, which increases the number of components. In the case of installing a heating coil on a flexible substrate, component cost of the flexible board is high and the flexible substrate needs to be configured separately from the heating coil. Therefore, there may be a problem that reliability of the heating coil cannot be guaranteed.

[0047] The present disclosure may provide an induction heating cooker capable of improving accuracy in detecting a position of a heating object by suppressing a misalignment in position of a sensor coil with respect to a heating coil. The present disclosure may provide an induction heating cooker capable of reducing the number of components. Hereinafter, embodiments of an induction heating cooker according to the present disclosure will be described with reference to the drawings.

[0048] The present disclosure relates to an induction heating cooker that inductively heats a heating object placed on a top plate, for example, a cooking tool such as a cooking pot. The present disclosure relates to an induction heating cooker capable of heating a heating object placed at any position on the top plate. FIG. 1 is a schematic diagram showing an entire configuration of an induction heating cooker according to an embodiment of the present disclosure. Referring to FIG. 1, an induction heating cooker **100** may include a top plate **1** on which a heating object is placed, and a substrate B. The substrate B may include a heating coil **2** and a sensor coil **3**. The heating coil **2** may inductively heat the heating object. The sensor coil **3** may detect a position of the heating object. The heating coil **2** and the sensor coil **3** may be integrated into one body. ‘Integrated’ means that the heating coil **2** and the sensor coil **3** form a structure that can be handled as a single assembly unit in a process of manufacturing the induction heating cooker **1**. In the current embodiment, the structure including the heating coil **2** and the sensor coil **3** may be implemented by the substrate B. An inverter circuit **4** may supply alternating current to the heating coil **2**. A control device **5** may control the inverter circuit **4**.

[0049] An outer surface of the top plate **1** may be a flat mounting surface on which a heating object is placed. The top plate **1** may be a flat plate formed of an electrical insulating material such as, for example, glass or ceramic.

[0050] The heating coil **2** may be installed on a bottom surface of the top plate **1**. Referring to FIG. 1, a plurality of heating coils **2** may be arranged to form a two-dimensional array (vertical and horizontal matrix) in a plan view. For example, as shown in FIG. 1, a pair of heating coils **2** may be installed on a common substrate B positioned on the bottom surface of the top plate **1**, and substrates B on each of which a pair of heating coils **2** are installed may be arranged to form a two-dimensional array (vertical and horizontal matrix) in a plan view.

[0051] The heating coil **2** may be installed in a sheet shape on the substrate B. For example, the heating coil **2** may be formed on the substrate B by a patterning process using photoresist, etc. In the embodiment shown in FIG. 1, two heating coils **2** may be installed on one substrate B. A distance between the two heating coils **2** may be smaller than half an outer diameter (for example, a maximum or minimum external dimension) of each heating coil **2**. The plurality of heating coils **2** may have the same shape and size. The plurality of heating coils **2** may not necessarily have the

same shape and size. For example, at least one of a shape or size of at least one of the plurality of heating coils **2** may be different from that of the others.

[0052] The sensor coil **3** may be installed on the bottom surface of the top plate **1**. For example, the sensor coil **3** may include an inductive proximity coil installed on the substrate B. The sensor coil **3** may detect a position of a heating object placed on the top plate **1**. FIG. **2** is a schematic diagram of an embodiment of a position detection sensor **30**. Referring to FIG. **2**, the sensor coil **3** may form the position detection sensor **30** together with a resonance capacitor **31**, an oscillation amplifier **32**, an oscillation frequency detection circuit **33**, and a controller **34**.

[0053] FIGS. **3** and **4** are schematic diagrams of an embodiment of the substrate B. Referring to FIGS. **3** and **4**, the substrate B may include a heating coil layer B1 including the heating coil **2**, and a sensor layer B2 including the sensor coil **3**. An insulating layer B3 may be positioned between the heating coil layer B1 and the sensor layer B2. In other words, the substrate B may be formed by laminating the heating coil layer B1, the insulating layer B3, and the sensor layer B2 into one body. Accordingly, the heating coil **2** and the sensor coil **3** may be integrated into one body and arranged, and a position of the sensor coil **3** with respect to the heating coil **2** may be set. That is, a relative positional relationship between the heating coil **2** and the sensor coil **3** in each substrate B may be common in the plurality of substrates B arranged.

[0054] Referring to FIGS. **3** and **4**, the sensor layer B2 may be positioned toward the top plate **1**, and the heating coil layer B1 may be positioned to an opposite side of the top plate **1** with respect to the sensor layer B2. On the heating coil layer B1, a connection connector CN connected to a control board **6** may be mounted. The connection connector CN may be mounted on the heating coil layer B1 in a direction opposite to the top plate **1**. By the connection connector CN, the substrate B may be connected to the control board **6**. That is, the heating coil **2** may be connected to the connection connector CN and connected to the control board **6** through the connection connector CN.

[0055] Also, the induction heating cooker **100** according to an embodiment may include a temperature sensor T as shown in FIGS. **1**, **3**, and **4**. The temperature sensor T may be installed on the substrate B or may be installed between neighboring substrates B. The temperature sensor T may detect a temperature of a heating object and/or the heating coil **2**. In the current embodiment, the temperature sensor T may be integrated into the heating coil **2** and the sensor coil **3**. The temperature sensor T may be positioned at or near a center of the heating coil **2**, or near the connection connector CN for connecting the heating coil **2** to the control board **6**.

[0056] The inverter circuit **4** may convert an alternating current voltage supplied from a power supply not shown into an arbitrary driving frequency and output the arbitrary driving frequency to the heating coil **2**. For example, the inverter circuit **4** may be a half-bridge inverter circuit using a switching element, or may be a full-bridge inverter circuit.

[0057] FIG. **5** is a functional block diagram of an embodiment of the control device **5**. Referring to FIGS. **1** and **5**, the control device **5** may include a processor **5a**, for example, a central processing unit (CPU), and memory **5b**. The control device **5** may further include an input device. The processor **5a** may read a program stored in the memory **5b** and execute the program. Accordingly, the control device **5** may function as a position detector **51** and an inverter controller **52**, as shown in FIG. **5**.

[0058] The position detector **51** may detect a position of a heating object based on an output value from the position detection sensor **30**. For example, the position detector **51** may detect a position and size of a heating object placed on the top plate **1**. For example, the position detection sensor **30** may output an inductance value of the sensor coil **3**. The inductance value may change depending on whether a heating object is placed above the sensor coil **3**.

[0059] An inductance value output from the position detection sensor **30** may change depending on a distance from a heating object to the sensor coil **3**. For example, when the heating object moves from a position far from the sensor coil **3** to a position close to the sensor coil **3**, the inductance

value may be reduced. The position detector **51** may detect a position and size of the heating object by comparing the inductance value output from the position detection sensor **30** to a preset threshold value. The position detector **51** may determine that the heating object is placed above a sensor coil **3** having an inductance value that is less than the threshold value, and output an upper side of the corresponding sensor coil **3** as a detection position at which the heating object is placed. [0060] The inverter controller **52** may calculate real power, which is actual power supplied from the power supply not shown, and control a driving frequency of the inverter circuit **4** such that the calculated real power comes close to target power corresponding to thermal power set by a user. The inverter controller **52** may control an on/off duty ratio of a switching device of the inverter circuit **4**. The inverter controller **52** may control the inverter circuit **4** to supply power selectively to a heating coil **2** located below a heating object placed on the top plate **1** among the plurality of heating coils **2**.

[0061] FIG. **6** is a view for describing arrangements of the sensor coil **3** with respect to the heating coil **2**. Arrangements of the sensor coil **3** with respect to the heating coil **2** will be described with reference to FIG. **6**. In FIG. **6**, an area where oblique lines are drawn represents a position where a heating object is placed.

[0062] First, a case in which one sensor coil **3** is positioned for one heating coil **2** is considered. For example, referring to FIG. **6(a)**, four heating coils **2** and four sensor coils **3** respectively corresponding to the four heating coils **2** are shown. In the case of this type of arrangement, because the number of the sensor coils **3** is too small, it may be difficult to ensure accuracy in detecting a position of a heating object. In order to ensure accuracy in detecting a position of a heating object, as shown in FIG. **6(b)**, a plurality of sensor coils **3**, for example, four sensor coils **3** may be arranged for one heating coil **2**. In this case, the number of components may increase, which may increase cost of the induction heating cooker. As shown in FIG. **6(c)**, two sensor coils **3** may be arranged for one heating coil **2** in a horizontal direction. In this case, the number of components may be less than in the arrangement shown in FIG. **6(b)** but it may be difficult to ensure accuracy of position detection in a vertical direction. As shown in FIG. **6(d)**, two sensor coils **3** may be arranged for one heating coil **2** in the vertical direction. In this case, the number of components may be less than in the arrangement shown in FIG. **6(b)** but it may be difficult to ensure accuracy of position detection in the horizontal direction.

[0063] FIG. **7** is a schematic diagram showing an embodiment of an arrangement of the sensor coil **3** with respect to the heating coil **2**. Referring to FIG. **7**, two sensor coils **3** may be arranged for one heating coil **2**. The two sensor coils **3** may be arranged diagonally with a center **2C** of the heating coil **2** in between. In other words, when an area where the heating coil **2** is formed is divided into four areas by two imaginary lines **ZX** and **ZY** that pass through the center **2C** of the heating coil **2** and are orthogonal to each other, the two sensor coils **3** may be respectively positioned in two diagonal areas facing each other with respect to the center **2C** of the heating coil **2**. For example, the two sensor coils **3** may be arranged in a first quadrant **Q1** and a third quadrant **Q3** or in a second quadrant **Q2** and a fourth quadrant **Q4**.

[0064] FIG. **8** shows calculation results of eddy current loss according to arrangements of the sensor coil **3**. FIG. **9** is a schematic view showing an embodiment of an arrangement of the sensor coil **3**. As shown in FIG. **8(a)**, as the sensor coil **3** is closer to a corner portion of the heating coil **2**, eddy current loss may increase. As shown in FIG. **8(c)**, when the sensor coil **3** is positioned in the vertical direction with respect to the heating coil **2** (that is, when the sensor coil **3** is positioned on the imaginary line **ZY** (see FIG. **7**)), it may be difficult to ensure accuracy of position detection in the horizontal direction, as described with reference to FIG. **6(d)**.

[0065] Referring to FIGS. **8(b)** and **9**, two sensor coils **3** may be arranged diagonally on one heating coil **2** at positions spaced from the corner portions of the heating coil **2** toward the center of the heating coil **2**. In other words, two sensor coils **3** may be arranged between the corner portions of the corresponding heating coil **2** and the imaginary line **ZY** (see FIG. **7**). According to this

arrangement, accuracy of position detection may be ensured while suppressing an increase of the number of components, and eddy current loss may be reduced compared to the arrangement shown in FIG. 8(a). Although not shown in the drawings, the sensor coils **3** may be arranged between the corner portions of the corresponding heating coil **2** and the imaginary line ZX (see FIG. 7). [0066] Hereinafter, a laminated structure of the heating coil layer **B1** and the sensor layer **B2** will be described. FIG. **10** is a schematic view for describing a configuration of a substrate in an existing heating coil **2'** using litz wires. FIG. **11** is a schematic view showing a configuration of a substrate in the case of using the heating coil **2** according to an embodiment of the present disclosure.

[0067] First, a configuration in which the existing heating coil **2'** using litz wires in the shape of a coil is formed on a heating coil layer **B1'** is shown in FIG. **10**. An output of an inner circumferential portion of the heating coil **2'** becomes a high output of, for example, up to 1000 V, and therefore, a voltage difference between the heating coil layer **B1'** and a sensor layer **B2'** of a low voltage increases. As a result, a thick insulation layer **B3'** needs to be positioned between the heating coil layer **B1'** and the sensor layer **B2'**, which may limit thinning of a substrate **B'**.

[0068] The substrate **B** according to an embodiment of the present disclosure may include, as shown in FIGS. **11(a)** and **11(b)**, two or more heating coil layers **B1** in which the heating coil **2** is installed and which are insulated from each other, a sensor layer **B2** in which the sensor coil **3** is installed, and an insulation layer **B3** positioned between the heating coil layer **B1** and the sensor layer **B2**, wherein a heating coil layer **B1a** connected to the resonance capacitor **31** (see FIG. **2**) among the two or more heating coil layers **B1** may be positioned farther away from the sensor layer **B2** than another heating coil layer **B1b**. In other words, a plurality of heating coil layers **B1** may be laminated such that directions of current flowing through two adjacent heating coil layers **B1** are opposite to each other. The plurality of heating coil layers **B1** may be insulated from each other by a plurality of core layers **B4**. That is, each core layer **B4** may be positioned between two adjacent heating coil layers **B1**. The heating coil layer **B1a** of an output side among the plurality of heating coil layers **B1** may be positioned farther away from the sensor layer **B2** than the heating coil layer **B1b** of an input side.

[0069] For example, in the case of a two-layer structure in which current is reciprocated once along two heating coil layers **B1** as shown in FIG. **11(a)**, outputs of inner circumferential portions of the heating coil layers **B1** may be suppressed to about 500 V even with the same output as in a heating coil having the shape of litz wires described above. For example, in the case of a four-layer structure in which current is reciprocated two times along four heating coil layers **B1** as shown in FIG. **11(b)**, outputs of inner circumferential portions of the heating coil layers **B1** may be suppressed to about 250 V. As a result, an increase in thickness of the insulation layer **B3** positioned between the heating coil layer **B1**, for example, the heating coil **B1b** of the input side and the sensor layer **B2** may be suppressed, which may contribute to thinning of the substrate **B**.

[0070] Also, a high-voltage part of the heating coil **2** may be applied to a through hole **TH** connecting the heating coil layer **B1**, the sensor layer **B2**, and the insulation layer **B3**, and therefore, the sensor coil **3** of the sensor layer **B2** may be installed without overlapping with the part. Accordingly, an increase in thickness of the insulation layer **B3** may be suppressed, which may contribute to thinning of the board **B**.

[0071] In the induction heating cooker **100**, the heating coil **2** and the sensor coil **3** may be integrated into one body, thereby suppressing a misalignment in position of the sensor coil **2** with respect to the heating coil **2**. Also, because the heating coil **2** having the sheet shape is used, the number of components may be reduced compared to the existing heating coil using litz wires. According to the configuration in which two sensor coils **3** are arranged diagonally with the center **2C** of the heating coil **2** in between, it is possible to improve accuracy in detecting a position of a heating object by using a small number of sensor coils **3**. According to the configuration in which a distance between two heating coils **2** installed on one substrate **B** is smaller than half an outer

diameter of each heating coil **2**, heating efficiency may be ensured while suppressing influence of the heating coils **2** on the sensor coil **3**. According to the configuration in which the heating coil layer **B1a** of the output side connected to the resonance capacitor **31** (see FIG. **2**) among the plurality of heating coil layers **B1** laminated is positioned farther away from the sensor layer **B2** than the heating coil layer **B1b** of the input side, the heating coil layer **B1b** close to the sensor layer **B2** may be designed with a low withstand voltage, which may enable a thin design of the insulation layer **B3**. According to the structure in which the temperature sensor **T** is integrated into the heating coil **2** and the sensor coil **3**, product costs may be reduced due to a reduction of wirings, etc. According to the structure in which the temperature sensor **T** is positioned at or near the center of the heating coil **2**, or near the connection connector **CN** to which the heating coil **2** is connected, the temperature sensor **T** may be positioned at a location where a temperature is most likely to be high to detect a temperature of the heating coil **2** or the connection connector **CN**, which enables control of the heating coil **2** with high reliability. According to the structure in which the connection connector **CN** is mounted on the substrate **B** in the direction opposite to the top plate **1**, a distance between a heating object and the heating coil **2** may be smaller than in a structure in which the connection connector **CN** is mounted toward the top plate **1**, and therefore, the heating object may be efficiently heated.

[0072] A configuration of the induction heating cooker **1** according to the present disclosure are not limited to the above-described embodiments. For example, in the above-described embodiments, two heating coils **2** may be positioned on one substrate **B**. However, one heating coil **2** may be positioned on one substrate **B**, or three or more heating coils **2** may be positioned on one substrate **B**. In the above-described embodiments, two sensor coils **3** may be positioned for one heating coil **2**. However, three or more sensor coils **3** may be positioned for one heating coil **2**. Also, in the above-described embodiments, the sensor coil **3** may be spaced from the corner portion of the heating coil **2** toward the center, but the sensor coil **3** may be positioned at the corner portion of the heating coil **2**. According to this arrangement, accuracy in detecting a position of a heating object in the horizontal and vertical directions may be improved even though a small number of sensor coils **3** are arranged.

[0073] FIG. **12** is a circuit diagram showing a configuration of the sensor coil **3**, a driving circuit portion **3D**, a MUX portion **36**, and a protection circuit portion **35** according to an embodiment of the present disclosure. Referring to FIG. **12**, the induction heating cooker **100** according to an embodiment may include the protection circuit portion **35**. The protection circuit portion **35** may be positioned between the sensor coil **3** and the driving circuit portion **3D** connected to the sensor coil **3**. The protection circuit portion **35** may protect the driving circuit portion **3D** from an induced voltage (for example, up to +80 V) generated in the sensor coil **3**.

[0074] The driving circuit portion **3D** may be provided with the resonance capacitor **31**, the oscillation amplifier **32**, the oscillation frequency detection circuit **33**, and the controller **34**, as shown in FIG. **2**. A plurality of sensor coils **3** may be connected to one driving circuit portion **3D** via multiplexers (MUX portions) **36** in between. The protection circuit portion **35** may be installed between each sensor coil **3** and a MUX portion **36** connected to the sensor coil **3**. That is, the protection circuit portion **35** may be installed to correspond to each of the plurality of sensor coils **3**.

[0075] For example, the protection circuit portion **35** may include a protection element **351** including a diode **351a** and a field effect transistor (FET) **351b**. The diode **351a** may include a schottky diode. The FET **351b** may include a metal oxide semiconductor field effect transistor (MOSFET). The protection element **351** may be connected in series to each of both terminals of the sensor coil **3**. In the protection element **351**, a drain of the FET **351b** may be connected to the sensor coil **3**. In the protection element **351**, the diode **351a** may be connected to a source of the FET **351b**. A cathode of the diode **351a** may be connected toward the sensor coil **3**, that is, to a source line of the FET **351b**, and an anode of the diode **351a** may be grounded. In FIG. **12**, the

protection element **351** may be configured such that, when a source voltage of the FET **351b** rises to 5 V or higher, the FET **351b** is automatically turned off.

[0076] FIG. **13** schematically shows operation states of an embodiment of the protection circuit portion **35**. FIG. **13(a)** shows a case in which the protection circuit portion **35** is in an off state (FET **351b** is on), and FIG. **13(b)** shows a case in which the protection circuit portion **35** is in an on state (FET **351b** is off). FIG. **13(b)** shows a state in which a source voltage of the FET **351b** becomes 5 V and the FET **351b** is turned off due to an induced voltage generated in a sensor coil **3** not selected by the MUX portion **36**. In addition, the case in which the protection circuit portion **35** is in an on state (FET **351b** is off) may include, for example, a case in which a source voltage of the FET **351b** becomes 5V due to an induced voltage generated in the sensor coil **3** when the MUX portion **36** is disable (invalid), and a case in which a source voltage of the FET **351b** becomes 5 V due to an induced voltage generated in the sensor coil **3** when an error is generated in the MUX portion **36**, the driving circuit portion **3D**, etc.

[0077] By the protection circuit portion **35**, the MUX portion or the driving circuit portion **3D** may be protected from an induced voltage generated in the sensor coil **3**. Because the protection circuit portion **35** described above protects the sensor coil **3** with a withstand voltage of the FET **351b** without consuming any induced voltage, the protection circuit portion **35** may also respond to a high induced voltage.

[0078] In the induction heating cooker **100** according to an embodiment, an output value (count value) of the sensor coil **3** may be corrected by using a temperature detected by the temperature sensor **T**. Here, resistance R_p of a resonance circuit may be expressed by Equation (1) below. In Equation (1), L is inductance of the sensor coil **3**, R_s is resistance of the sensor coil **3**, and C is capacitance of the resonance capacitor **31**.

$$R_p = L / (R_s \times C) \quad \text{Equation (1)}$$

[0079] The driving circuit portion **3D** that drives the sensor coil **3** may control the sensor coil **3** with constant current and output a ratio of frequencies by a change of an inductance value as a count value. That is, the count value may be proportional to resistance R_p of the resonance circuit. A change of resistance R_p of the resonance circuit according to a temperature may be expressed by Equation (2) below when a coil material of the sensor coil **3** is constant. α is a temperature coefficient of resistance.

$$R_p(t) = R_p(t_0) \times (1 + \alpha(t - t_0)) \quad \text{Equation (2)}$$

[0080] That is, the count value may decrease linearly with respect to a temperature change. Accordingly, the count value may be corrected based on, for example, Equation (3) below by using a temperature change ($T - T_0$) with respect to a preset reference temperature T_0 and a temperature correction coefficient k obtained in advance.

$$\text{Count Value after Correction} = \text{Count Value before Correction} + k \times (T - T_0) \quad \text{Equation (3)}$$

[0081] Correction of a count value according to a temperature may be performed by the oscillation frequency detection circuit **33** or the controller **34** of the position detection sensor **30** or by the position detector **51** of the control device **5**. FIG. **14** shows a count value in the case in which no temperature correction is performed and a count value in the case in which temperature correction is performed. FIG. **14(a)** shows a change of a detection value in the case in which no temperature correction is performed. In this case, a count value decreases after heating starts, and even though there is a pot, the count value falls below a threshold value for determining that “there is no pot”, resulting in false detection. Meanwhile, FIG. **14(b)** shows a change of a detection value in the case in which temperature correction is performed. In this case, a decrease of a count value is suppressed even after heating starts, thereby preventing false detection.

[0082] The induction heating cooker **100** according to an embodiment may compensate a change of

an output value (inductance value) of the sensor coil 3, which occurs due to magnetic flux absorption by the heating coil 2 when the heating coil 2 starts to operate. For example, when the heating coil 2 starts to operate, an amount of change of an output value (inductance value) of the sensor coil 3 caused due to magnetic flux absorption by the heating coil 2 may be measured, and the measured amount of change may be added to an output value (inductance value) of the sensor coil 3 after the heating coil 2 starts to operate. In this way, a change of an output value (inductance value) of the sensor coil 3 caused due to magnetic flux absorption by the heating coil 2 may be compensated. Correction of an output value with respect to influence by magnetic flux absorption may be performed by the oscillation frequency detection circuit 33 or the controller 34 of the position detection sensor 30 or by the position detector 51 of the control device 5.

[0083] FIG. 15 shows a count value in the case in which magnetic flux absorption is not corrected and a count value in the case in which magnetic flux absorption is corrected. FIG. 15(a) shows a change of a detection value in the case in which magnetic flux absorption is not corrected. In this case, a count value decreases due to magnetic flux absorption by the heating coil 2 after heating starts. Meanwhile, FIG. 15(b) shows a change of a detection value in the case in which magnetic flux absorption is corrected. In this case, a decrease of a count value caused due to magnetic flux absorption by the heating coil 2 is compensated.

[0084] An induction heating cooker according to an aspect of the present disclosure may include a top plate, and a substrate positioned on a bottom surface of the top plate. On the substrate, a heating coil for inductively heating a heating object may be provided in a sheet shape. A sensor coil for detecting a position of the heating object may be integrated into the heating coil and provided on the substrate. By this configuration, because the heating coil and the sensor coil are integrated into one body, a misalignment in position of the sensor coil with respect to the heating coil may be suppressed. Also, because the heating coil having a sheet shape is used, the number of components may be reduced compared to an existing heating coil using litz wires.

[0085] As an example, at least two sensor coils may be arranged for one heating coil. The two sensor coils may be arranged diagonally with a center of the heating coil in between. Accordingly, accuracy in detecting a position of a heating object may be improved even though a small number of sensor coils are used.

[0086] As an embodiment, the sensor coil may be spaced from a corner portion of the heating coil toward the center of the heating coil. As an embodiment, the sensor coil may be positioned between the corner portion of the heating coil and two imaginary lines that pass through the center of the heating coil and are orthogonal to each other. Accordingly, eddy current loss may be reduced.

[0087] As an embodiment, at least two heating coils may be arranged on the substrate. A distance between the two heating coils 2 may be smaller than half an outer diameter of each heating coil. Accordingly, heating efficiency may be ensured while suppressing influence on the sensor coils by the heating coil.

[0088] As an embodiment, the substrate may include two or more heating coil layers each including the heating coil and insulated from each other; a sensor layer including the sensor coil; and an insulation layer positioned between the heating coil layers and the sensor layer. The sensor layer may be positioned closer to the top plate than the heating coil layers. As an embodiment, a heating coil layer connected to a resonance capacitor among the two or more heating coil layers may be positioned farther away from the sensor layer than another heating coil layer. Accordingly, the heating coil layer closer to the sensor layer may be designed with a low withstand voltage, which may enable a thin design of the insulation layer.

[0089] As an embodiment, the induction heating cooker may include a temperature sensor configured to detect a temperature of at least one of the heating coil or the heating object and integrated into the heating coil and the sensor coil. By integrating the temperature sensor into the heating coil and the sensor coil, product costs may be reduced due to a reduction in wirings, etc.

[0090] As an embodiment, the temperature sensor may be positioned at the center of the heating

coil, near the center of the heating coil, or near a connection connector to which the heating coil is connected. Because the temperature sensor is positioned at a location where a temperature is most likely to be high, a temperature of the heating coil or the connection connector may be detected, which enables control on the heating coil with high reliability.

[0091] As an embodiment, the connection connector for connecting the heating coil to the control board may be provided on the substrate. The connection connector may be mounted on the substrate in a direction opposite to the top plate. As an embodiment, the connection connector may be provided on the heating coil layer of the substrate, on which the heating coil is provided. Accordingly, a distance between the heating object and the heating coil may be smaller than in a structure in which the connection connector is toward the top plate, and accordingly, the heating object may be efficiently heated.

[0092] As an embodiment, the induction heating cooker may include a driving circuit portion connected to the sensor coil; and a protection circuit portion positioned between the sensor coil and the driving circuit portion and configured to protect the driving circuit portion from an induced voltage generated in the sensor coil. As an embodiment, a plurality of sensor coils may be connected to the driving circuit portion, and the protection circuit portion may be provided for each of the plurality of sensor coils.

[0093] As an embodiment, the induction heating cooker may include a temperature sensor T configured to detect a temperature of at least one of the heating coil or the heating object. The induction heating cooker may correct an output value of the sensor coil by using a temperature detected by the temperature sensor. The output value of the sensor coil may change by being influenced by a temperature around the sensor coil. More specifically, when resistance of the sensor coil changes depending on a temperature around the sensor coil, the output value of the sensor coil may change. For example, when a temperature around the sensor coil rises, the output value of the sensor coil may decrease. In this case, the output value of the sensor coil may decrease to the same state as “there is no pot” during heating, and false detection of “there is no pot” may occur, which may result in stopping heating. By correcting an output value of the sensor coil by using a temperature detected by the temperature sensor, false detection and stopping heating due to such false detection may be prevented.

[0094] As an embodiment, the induction heating cooker may compensate a change of an output value of the sensor coil caused due to magnetic flux absorption by the heating coil when the heating coil starts to operate. While the heating coil does not operate, the sensor coil may not be influenced by the heating coil because a resonance circuit including the heating coil is not configured. Meanwhile, while the heating coil operates, a resonance circuit including the heating coil may be configured, and magnetic flux generated in a sensor coil near the corresponding heating coil may be absorbed in the heating coil. As a result, there may be possibility that accuracy in detecting a position of a heating object will deteriorate. By compensating a change of an output value of the sensor coil caused due to magnetic flux absorption by the heating coil when the heating coil starts to operate, it may be possible to prevent accuracy in detecting a position of a heating object from deteriorating.

[0095] Although the induction heating cooker according to the disclosure have been described in connection with the limited embodiments and the drawings, the disclosure is not limited to the embodiments, and various modifications may be made without departing from the purpose.

Claims

1. An induction heating cooker comprising: a top plate; a substrate positioned on a bottom surface of the top plate; at least one heating coil on the substrate and configured to inductively heat a heating object, the at least one heating coil being sheet-shaped; and at least one sensor coil configured to detect a position of the heating object, wherein the at least one sensor coil is on the

substrate so that the at least one heating coil and the at least one sensor coil are integrated into a single body.

2. The induction heating cooker of claim 1, wherein the at least one sensor coil includes at least two sensor coils arranged for the at least one heating coil, and the at least two sensor coils are arranged diagonally about a center of the at least one heating coil.
3. The induction heating cooker of claim 2, wherein each sensor coil of the at least two sensor coils is spaced apart from a corner portion of the at least one heating coil toward the center of the at least one heating coil.
4. The induction heating cooker of claim 2, wherein each sensor coil of the at least two sensor coils is positioned between a corner portion of the at least one heating coil and two orthogonal lines that pass through the center of the at least one heating coil.
5. The induction heating cooker of claim 1, wherein the at least one heating coil includes at least two heating coils arranged on the substrate, and a distance between the at least two heating coils is smaller than half an outer diameter of each heating coil of the at least two heating coils.
6. The induction heating cooker of claim 1, wherein the at least one heating coil includes at least two heating coils arranged on the substrate, the substrate includes: two or more heating coil layers each including at least one heating coil of the at least two heating coils, the two or more heating coil layers being insulated from each other, a sensor layer including the at least one sensor coil, and an insulation layer positioned between the two or more heating coil layers and the sensor layer, and the sensor layer is positioned closer to the top plate than are the two or more heating coil layers.
7. The induction heating cooker of claim 6, wherein one heating coil layer of the two or more heating coil layers is connected to a resonance capacitor, and is positioned farther away from the sensor layer than is another heating coil layer of the two or more heating coil layers.
8. The induction heating cooker of claim 1, further comprising: a temperature sensor configured to detect a temperature of at least one of the at least one heating coil or the heating object, and integrated into the single body.
9. The induction heating cooker of claim 8, wherein the temperature sensor is positioned at a center of the at least one heating coil, near the center of the at least one heating coil, or near a connection connector to which the at least one heating coil is connected.
10. The induction heating cooker of claim 1, further comprising: a control board to control the at least one heating coil; and a connection connector connecting the at least one heating coil to the control board, and provided on the substrate, wherein the connection connector is mounted on the substrate in a direction opposite to the top plate.
11. The induction heating cooker of claim 10, wherein the connection connector and the at least one heating coil are on a heating coil layer of the substrate.
12. The induction heating cooker of claim 1, further comprising: a driving circuit portion connected to the at least one sensor coil; and at least one protection circuit portion positioned respectively between the at least one sensor coil and the driving circuit portion, and configured to protect the driving circuit portion from an induced voltage generated in the at least one sensor coil.
13. The induction heating cooker of claim 12, wherein the at least one sensor coil includes a plurality of sensor coils, the at least one protection circuit portion includes a plurality of protection circuit portions, the plurality of sensor coils are connected to the driving circuit portion, and one protection circuit portion of the plurality of protection circuit portions is provided for each sensor coil of the plurality of sensor coils.
14. The induction heating cooker of claim 1, further comprising: a temperature sensor configured to detect a temperature of at least one of the at least one heating coil or the heating object, wherein the induction heating cooker is configured to correct an output value of the at least one sensor coil by using the temperature detected by the temperature sensor.
15. The induction heating cooker of any claim 1, wherein the induction heating cooker is configured to compensate for a change of an output value of the at least one sensor coil caused due

to magnetic flux absorption by the at least one heating coil when the at least one heating coil starts to operate.
